Online Resource 4 – Summary of primary studies included in photovoice meta-analyses

Article title: Identifying evidence of the effectiveness of photovoice: a systematic review and meta-analysis of the international healthcare literature
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REFERENCE	POPULATION, SETTING(S)	STUDY DESIGN	INTERVENTION GROUP, *N* FOR RELEVANT OUTCOME(S)	COMPARISON/CONTROL GROUP, *N* FOR RELEVANT OUTCOME(S)	EVALUATING PARTICIPATION IN PV ACTIVITIES, OR SIMPLY VIEWING OF PV MATERIALS?	DURATION (RELATED TO THOSE INVOLVED IN EVALUATION)	RELEVANT OUTCOME(S)	REPORTED POST-INTERVENTION DATA, INTERVENTION – COMPARISON
ARIYO & OLOYEDE 2017	University of Ibadan (Nigeria) undergraduate students	Cross-sectional	Photo-aided 24 hour dietary recall (n=200)	Conventional 24 hour dietary recall (n=200)	Participation	24 hours	Recall of nutrient intake (combined)	P-value of t-test = 0.7272
BAIG ET AL 2015	Latino adults with diabetes in Chicago (US)	RCT	A church-based diabetes self-management programme led by trained lay leaders that included 90-min group classes and a photovoice exercise (n=43)	Usual care with one 90-minute lecture on diabetes self-management (n=41)	Participation	8 weeks	Glycosylated haemoglobin (%) 3-m Glycosylated haemoglobin (%) 6-m Low density lipoprotein (mg/dL) 3-m Low density lipoprotein (mg/dL) 6-m Systolic blood pressure (mmHg) 3-m Systolic blood pressure (mmHg) 6-m Diastolic blood pressure (mmHg) 3-m Diastolic blood pressure (mmHg) 6-m Waist circumference 3-months Waist circumference 6-months Diabetes self-empowerment score 6-m Participating in ≥30 min physical activity 6-m Following a healthful eating plan 6-m Testing blood sugar as recommended 6-m Taking insulin injections 6-m Taking oral medications 6-m	Mean change = −0.28 (−0.70,0.14) vs −0.36 (−0.79,0.07) Mean change = −0.27 (−0.81,0.28) vs −0.05 (−0.59,0.48) Mean change = −0.16 (−6.61,6.30) vs 0.97 (−5.78,7.72) Mean change = −4.94 (−13.64,3.77) vs 2.55 (−6.04,11.13) Mean change = −2.72 (−7.19,1.74) vs 1.42 (−3.29,6.13) Mean change = −4.62 (−9.14,−0.09) vs 0.60 (−3.85,5.04) Mean change = −4.07 (−7.19,−0.94) vs 1.78 (−1.50,5.06) Mean change = −3.14 (−6.59,0.31) vs 0.91 (−2.50,4.33) Mean change = −0.93 (−2.63,0.76) vs −0.22 (−1.99,1.55) Mean change = −0.98 (−2.83,0.88) vs −0.51 (−2.34,1.32) Mean (SD) = 4.11 (0.63) – 4.05 (0.48) Mean (SD) = 3.78 (2.88) – 4.00 (2.65) Mean (SD) = 3.80 (2.78) – 4.73 (2.74) Mean (SD) = 3.95 (2.85) – 4.05 (2.89) Mean (SD) = 6.86 (0.53) – 7.00 (0) Mean (SD) = 6.77 (1.19) – 6.86 (0.85)
BLUMENSTOCK ET AL 2015	Middle School students with asthma in Chicago (US)	Pre-posttest	Mini-tablets to take pictures of socio-environmental factors influencing their asthma, recording video and co-creating a targeted community intervention	Pre-survey: n=12	Participation	13 weeks	Asthma Belief Survey Sociopolitical Control Scale	P-value of t-test = 0.03 P-value of t-test = 0.033

			(post-survey: n=12)					
CARROLL ET AL 2018	Indigenous Cherokee people in Oklahoma (US)	Pre-posttest	Photovoice in collaboration with elders who supply cultural guidance to the Cherokee Nation state government. Still-image documentary developed highlighting themes photographed by the elders, with survey before and after viewing this (post-survey: n=84)	Pre-survey: n=84	Viewing	1 hour	"How much health of your community where you live is affected by the condition of the surrounding natural environment" "I am concerned about local environmental issues in the Cherokee Nation"	Percentage answered "a lot": 78% – 73% of 84 participants Percentage answered "strongly agree": 50% – 45% of 84 participants
CHAN ET AL 2015	Young adults, mainly Hispanic, in a low-income primary care clinic in Southern California (US)	Pre-posttest	A fotonovela (photographic short story) intervention to improve human papillomavirus vaccination knowledge, attitudes, and intention (post-survey: n=38)	Pre-survey: n=38	Viewing	25 minutes	Perceived susceptibility Perceived benefit of vaccination Intent to vaccinate Intent to encourage others to vaccinate	Number answered 'likely': 12 – 8 Number answered 'likely': 33 – 30 Number answered 'likely': 31 – 24 Number answered 'likely': 34 – 30
FLANAGAN ET AL 2016	Primary care providers in New Haven (US)	RCT	Intervention to reduce stigma towards people with mental illness through a one-hour photovoice performance followed by a discussion	Did not attend the performance (n=13)	Viewing	1 hour	Responsibility Dangerousness Anger Fear Social Distance Competence Assessment Instrument	Mean (SE) = 2.1 (0.2) – 2.5 (0.2) Mean (SE) = 2.3 (0.2) – 3.2 (0.3) Mean (SE) = 2.1 (0.2) – 2.2 (0.2) Mean (SE) = 1.8 (0.1) – 2.8 (0.1) Mean (SE) = 3.5 (0.2) – 4.0 (0.2) Mean (SE) = 3.4 (0.0) – 3.2 (0.0)

			(n=14)					
GOZALIANS 2012	Youth aged 11 to 18 years old from South Los Angeles, California (US)	Pre-posttest	Took part in photovoice intervention with training and discussions of change (post-survey: n=133)	Pre-survey: n=133	Participation	8 weeks	Self-empowerment (healthy eating)	P-value of t-test = 0.316
GUPTA ET AL 2013	Students in an urban, Chicago high school (US), including with asthma and/or household member(s) with asthma	Pre-posttest	Photographs from a photovoice project integrated into public service announcements to increase community asthma awareness (Post-survey: n=89)	Pre-survey: n=89	Viewing	3 minutes	Asthma definition: Lungs Asthma definition: Inflammation	Correct response (%) = 98.9 – 47.2 Correct response (%) = 62.9 – 9.0
JEONG ET AL 2012	Clinical trial volunteers from Chungnam National University Hospital Regional Clinical Trial Center (South Korea)	RCT	Informed consent for clinical studies with photonovella information sheets, developed based on participant observation and piloted on matched participants (n=50)	Informed consent with standard paper information sheets (n=50)	Viewing	1 hour	Objective understanding choice type Objective understanding short answer type	Mean (SD) = 15.9 (2.3) – 16.3 (2.6) Mean (SD) = 15.5 (2.6) – 15.9 (3.2)
JIA ET AL 2019	African-American population from diabetes 'hotspots' in Boston, Massachusetts (US)	Pre-posttest	Six photovoice-community health worker group sessions at a public library (Post-survey: n=15)	Pre-survey: n=15	Participation	12 weeks	Diabetes self-management (post) Diabetes self-management (3 months' follow-up) Level of stress (3 months' follow-up)	Mean (SD) = 31.9 (6.3) – 34.5 (7.0) Mean (SD) = 34.4 (6.5) – 34.5 (7.0) Mean (SD) = 5.7 (3.8) – 6.4 (2.1)

KILANOWSKI 2014	Latino/a migrant students in the Midwest (US)	Pre-posttest	Peer-education health and safety education program with student-produced videos and photovoice (Post-survey: n=94)	Pre-survey: n=108	Participation (middle school students) and viewing (lower school students)	7 weeks	Total safety score Injury experience	Median(IQR),N=29.0(10–44),94–30.0(7–53),107 Median(IQR),N = 0(0–4),94 – 0(0–4),108
LEE ET AL 2013	Chinese, Korean, and Vietnamese American Communities in Maryland (US)	Cross-sectional (originally RCT, but only relevant data from intervention group presented)	Culturally appropriate photonovella development and process for Hepatitis B prevention (n=347 that read the photonovella)	Control arm did not receive the photonovel (n=436)	Viewing	1 month	Intention to screen (n=235) Self-efficacy (n=322) (both results based on those that agreed that the "Story is written by someone who knows the community" to present results most relevant to photovoice as representing the community voice)	OR (95% CIs) = 2.2 (1.0, 4.4) OR (95% CIs) = 5.8 (3.3, 10.2)
LEUNG ET AL 2019	Older adults with diabetes and/or hypertension in community centres in the Western District of Hong Kong Special Administrative Region (China)	Non-randomised intervention study	Participants took photos of barriers and facilitators to physical activity and worked out a physical activity plan through a meeting supervised by a nurse (n=107)	Waitlist control (n=97)	Participation	10 weeks	Number of steps taken Self-efficacy for doing physical activities	Mean (SD) = 11686 (4594) – 11643 (4803) Median (IQR) = 5.22 (2.89) – 5.44 (2.78)
LEVY ET AL 2019	Female paediatric palliative caregivers in New York (US)	Pre-posttest	A meaning-making curriculum where photo narratives where created and shared via social media and photos were displayed in a final	Pre-survey: n=9	Participation	9 weeks	Presence of meaning	Mean (SD) = 31.86 (3.49) – 28.86 (4.14)

			exhibition (Post-survey: n=7)					
MORSE 2018	American Indian and Alaska Native people within Denver Indian Health and Family Services urban clinic (US)	RCT	Delivery of health information through Multimedia Messaging Service (MMS); text with images. Last message of week asked participants to respond with photo and text (n=11)	Short Message Service (SMS); health information delivered through text only and response from participants to last message of week only by text (n=12)	Participation	5 weeks	Heart Attack Knowledge Heart Health Knowledge Physical Activity Knowledge Nutrition Knowledge	ANOVA (condition) = 0.08 ANOVA (condition) = 3.28 ANOVA (condition) = 0.02 ANOVA (condition) = 0.67
OOSMAN 2012	First Nations and Métis school-aged children, Saskatchewan (Canada)	Non-randomised interventio n study	Intervention class (Grade 3-4 split class) with classroom lessons and activities related to health, physical activity, healthy eating and diabetes prevention, including photovoice elements (n=16)	Two other classes (independent Grade 3 and 4 class) acted as a standard care comparison group (n=21)	Participation	4 months	Curricular knowledge (physical activity, eating, diabetes) Food and physical activity self-efficacy Average minutes per day (moderate-vigorous activity)	Median (range) = 4.5 (3–6) – 4.0 (1–7) Median (range) = 27.0 (13–33) – 28.0 (21–33) Mean (SD) = 56 (23) – 50 (18)
OVERBY ET AL 2018	3 high schools from a Midwest state (US)	Cross-sectional	School wellness curriculum: 4 educational sessions, 1 intersession, and 1 photo exhibit (n=49 who had completed the full curriculum)	Strictly attended the photovoice exhibit (n=67)	Participation	A school year? Integrated as part of curriculum and held during classroom time	No change in awareness	Percentages: 2% of 49 – 13% of 67
PATIDAR ET AL 2012	Women affected by	Pre-posttest	Captured photographs of	Pre-survey: n=9	Participation	N/A	Depressive symptoms (Beck Depression Inventory-II) Social Connectedness Scale	Mean (SD) = 9.00 (3.25) – 10.25 (3.85) Mean (SD) = 42.13 (6.75) – 40.25 (8.81)

	ovarian cancer (US)		experience with cancer and participated in two photovoice group discussions (Post-survey: n=9)					
RUSSINOVA ET AL 2014	People with serious mental illnesses in Boston (US)	RCT	Antistigma peer-led and group-based photovoice programme (n=40)	Wait-list control group (n=42)	Participation	10 weeks	Internalized Stigma of Mental Illness Scale (post) Internalized Stigma of Mental Illness Scale (3-m) Personal Growth and Recovery Scale (post) Personal Growth and Recovery Scale (3-months) Center for Epide. Studies Depression Scale (post) Center for Epide. Studies Depression Scale (3-m) Generalized Perceived Self-Efficacy Scale (post) Generalized Perceived Self-Efficacy Scale (3-m) Empowerment Scale (post) Empowerment Scale (3-months) Community activism and autonomy (post) Community activism and autonomy (3-months) Hopefulness (post) Hopefulness (3-months)	Mean (SD) = 2.04 (0.42) – 2.13 (0.47) Mean (SD) = 2.01 (0.41) – 2.10 (0.45) Mean (SD) = 3.20 (0.54) – 3.08 (0.63) Mean (SD) = 3.33 (0.51) – 3.12 (0.59) Mean (SD) = 21.59 (12.65) – 22.29 (14.55) Mean (SD) = 22.60 (12.16) – 21.52 (14.77) Mean (SD) = 3.07 (0.60) – 2.92 (0.52) Mean (SD) = 3.04 (0.65) – 2.97 (0.62) Mean (SD) = 2.89 (0.32) – 2.84 (0.29) Mean (SD) = 2.89 (0.35) – 2.88 (0.33) Mean (SD) = 3.33 (0.46) – 3.18 (0.36) Mean (SD) = 3.36 (0.47) – 3.24 (0.42) Mean (SD) = 2.79 (0.43) – 2.70 (0.50) Mean (SD) = 2.71 (0.58) - 2.90 (0.44)
RUSSINOVA ET AL 2018	People with serious mental illnesses in Boston (US)	RCT	Peer-run photography-based group intervention (Vocational Empowerment Photovoice) (n=22)	Wait-list control group (n=26 post-intervention, n=25 at 3 months' follow-up)	Participation	10 weeks + 2 booster sessions a month apart after core curriculum	Internalized Stigma of Mental Illness Scale (post) Internalized Stigma of Mental Illness Scale (3-m) Empowerment Scale (post) Empowerment Scale (3-months) Self-esteem/self-efficacy (post) Self-esteem/self-efficacy (3-months) Community activism (post) Community activism (3-months) Hopefulness (post) Hopefulness (3-months) Work Hope Scale (post) Work Hope Scale (3-months) Vocational identity (post) Vocational identity (3-months) Work capacity (post) Work capacity (3-months) Work motivation (post) Work motivation (3-months)	Mean (SD) = 1.86 (0.44) – 2.23 (0.41) Mean (SD) = 1.92 (0.50) – 2.08 (0.43) Mean (SD) = 2.98 (0.34) – 2.92 (0.24) Mean (SD) = 2.96 (0.34) – 2.88 (0.29) Mean (SD) = 2.96 (0.45) – 2.91 (0.38) Mean (SD) = 2.90 (0.52) – 2.81 (0.45) Mean (SD) = 3.39 (0.45) – 3.32 (0.37) Mean (SD) = 3.41 (0.36) – 3.33 (0.40) Mean (SD) = 2.94 (0.52) – 2.82 (0.36) Mean (SD) = 3.00 (0.51) – 2.77 (0.34) Mean (SD) = 4.64 (1.29) – 4.29 (1.15) Mean (SD) = 4.67 (1.24) – 4.58 (0.94) Mean (SD) = 9.09 (5.08) – 6.77 (4.99) Mean (SD) = 9.23 (5.05) – 8.32 (5.17) Mean (SD) = 6.00 (2.66) – 6.23 (2.16) Mean (SD) = 6.14 (2.75) – 6.36 (2.63) Mean (SD) = 3.15 (0.58) – 3.09 (0.55) Mean (SD) = 3.17 (0.60) – 3.04 (0.50)
SEGURA-PÉREZ ET AL 2017	Adult Latin American population in Hartford, Connecticut	Pre-posttest	Reading a co-designed culturally appropriate heart disease	Pre-survey: n=100	Viewing	N/A	Knowledge on seven steps for heart disease prevention	Mean (SD) = 78.8 (13.1) – 63.7 (16.3)

	(US)		prevention fotonovela (Post-survey: n=100)					
SEVERCAN 2012	9-11 year-old children in Istanbul (Turkey)	Pre-posttest	Urban regeneration and involvement in planning and design activities including photovoice (Post-survey: n=123)	Pre-survey: n=282	Participation	1 year	Knowledge on social attributes of block Knowledge on social attributes of neighbourhood Knowledge on social attributes of city	F-test = 1.57 F-test = 3.12 F-test = 0.44
SHAYNA 2015	Women below the poverty line with diabetes in Detroit (Michigan, US)	Non-randomised interventio n study	Of 24 participants who took part in a national *Cooking Matters* course, the majority captured photos to describe changes in experience with food and diabetes as a result of the course (n=14)	Comparison group that did not participate in the *Cooking Matters* course nor captured any photos (n=24)	Participation	7 weeks	Comprehending nutrition facts, almost always or always Exercise, almost always or often Eating breakfast, almost always or always At least 2 cups of fruit daily, almost always or often At least 3 food groups each meal, almost always or often	Percentages: 67% of 24 – 29% of 24 Percentages: 87% of 24 – 59% of 24 Percentages: 75% of 24 – 42% of 24 Percentages: 50% of 24 – 38% of 24 Percentages: 80% of 24 – 54% of 24
SNOW 2012	Women of child bearing years located in Davis County, Salt Lake County and Weber County, Utah (US)	RCT	Photo novella lessons (n = 152)	Only existing lessons of the Expanded food and nutrition education program (n = 95)	Viewing	3 months (average of 11 lessons of 45 minutes each)	Listeriosis and preventive behaviour knowledge	Mean (SD) = 5.64 (2.05) – 5.89 (2.07)
TETI ET AL 2018	Patient viewers of posters that were placed in a clinic in a Midwestern	Pre-posttest	A group of people living with HIV used photovoice to develop 10 posters	Pre-survey of patient viewers of posters: n=57	Viewing	3 months	Brief Estimate of Health Knowledge and Action Self-efficacy Adherence to medication	Mean (SD) = 3.5 (0.8) – 3.4 (0.8) Mean (SD) = 91 (33) – 99 (27) Number (percent) = 16 (36%) – 19 (39%)

	city (US)		describing their medication adherence experiences. Patients in the clinic that viewed the posters responded to survey (post-survey: n=54)					
TIPPIN & MARANZAN 2019	Undergraduate students enrolled in psychology courses at a mid-size Canadian university	RCT	Viewed photovoice-based antistigma video developed over 9 months by people with mental illness (post n=156, 1-month follow-up n=56)	Viewed control video (post n=147, 1-month follow-up n=48)	Viewing	18 minutes	Responsibility (post) Responsibility (1 month follow-up) Anger (post) Anger (1 month follow-up) Fear (post) Fear (1 month follow-up) Dangerousness (post) Dangerousness (1 month follow-up) Social distance (post) Social distance (1 month follow-up)	Mean (SD) = 2.93 (1.56) – 2.92 (1.38) Mean (SD) = 2.79 (1.33) – 3.02 (1.22) Mean (SD) = 1.78 (1.26) – 2.17 (1.45) Mean (SD) = 1.96 (0.97) – 2.06 (1.21) Mean (SD) = 1.92 (1.42) – 2.32 (1.43) Mean (SD) = 1.96 (1.19) – 2.32 (1.36) Mean (SD) = 2.80 (1.11) – 2.98 (1.01) Mean (SD) = 2.64 (0.96) – 2.83 (1.01) Mean (SD) = 1.92 (0.68) – 2.09 (0.59) Mean (SD) = 0.81 (0.51) – 1.10 (0.49)
WOITH ET AL 2015	Health care workers in a regional tuberculosis hospitals in Russia	Pre-posttest	Photovoice task combined into a poster displayed at workplace (post-exhibition survey: n=20)	Pre-exhibition survey: n=20	Participation and viewing	6 months	Intention to wear respirators Intention to fit-check respirators	Mean (SD) = 3.75 (0.44) – 3.50 (0.61) Mean (SD) = 3.85 (0.37) – 3.60 (0.50)
YI-FRAZIER ET AL 2015	Adolescents with type 1 Diabetes (US)	Pre-posttest	Post images on Instagram (post-survey: n=13)	Pre-survey: n=16	Participation	3 months	Self-esteem	Mean (SD) = 26.8 (4.7) – 24.2 (5.1)

CI = confidence intervals; HIV = human immunodeficiency virus; IQR = interquartile range; OR = odds ratio; RCT = randomised controlled trial; SD = standard deviation; SE = standard error.